

Functions of many variables – 3

Maps of $\mathbb{R}^n$ into itself: the Jacobian, invertibility and rigidity

Homework

11 сентября 2026 года

Theoretical minimum. A non-vanishing Jacobian gives a local diffeomorphism and nothing more; to pass from local to global one needs properness (Hadamard) or monotonicity of the map. The implicit function theorem is the inverse function theorem for $(x, y) \mapsto (x, F(x, y))$, and it is the standard way to prove that an implicitly defined quantity depends smoothly on a parameter; when the hypotheses fail, the same conclusion may still be reachable by a compactness and uniqueness argument. A C^1 map with non-vanishing Jacobian is open, so a proper such map is surjective. Rigidity statements — spheres to spheres, circles to circles, one distance preserved — are proved by recovering midpoints, or lines, or equilateral configurations from the given family, and then upgrading additivity to linearity by continuity.

1. Prove the implicit function theorem from the inverse function theorem, and use it to show that a simple root of a polynomial depends smoothly on its coefficients while a multiple root need not depend on them even continuously in a differentiable way.
2. (МФТИ 2016) A function $f(x, y)$ is continuous on $[0, 1] \times [0, 1]$ and has the following property: for every fixed x the minimum of $f(x, y)$ over y is attained at exactly one point $y = g(x)$. Prove that the function g obtained in this way is continuous.
3. (Proposed problems) Suppose $n \geq 1$ and $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a map such that the image of the interior of every sphere of codimension 1 is the interior of a sphere of codimension 1 of the same radius. Prove that f is an isometry.

4. Prove that a C^1 map $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $\det Df(x) \neq 0$ everywhere is open, and deduce that if in addition $\|f(x)\| \rightarrow \infty$ as $\|x\| \rightarrow \infty$, then f is onto. Show by an example that without the second condition the image can be a proper open subset of $\mathbb{R}^n$.
5. (Mexmar, 2024) A continuous injective map $t \mapsto (x(t), y(t))$ from $[0, 1]$ to $\mathbb{R}^2$ satisfies $(x(t) - x(s))(y(t) - y(s)) \geq 0$ for all $t, s \in [0, 1]$. Prove that $t \mapsto x(t)$ is monotone.
6. (CIIM 2025) Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $F(x, y) = (x + y - x^3, y - y^3)$. Put $F^1 = F$ and $F^{n+1} = F \circ F^n$ for $n \geq 1$.
 - a) Prove that if $|x| \leq \frac{1}{2}$ and $|y| \leq \frac{1}{2}$, then $\lim_{n \rightarrow \infty} F^n(x, y) = (0, 0)$.
 - b) Prove that for any $K > 0$ and $\varepsilon > 0$ there exist a positive integer n and a point $(x_0, y_0) \in \mathbb{R}^2$ with $0 < \|(x_0, y_0)\| < \varepsilon$ such that $\|F^n(x_0, y_0)\| > K\|(x_0, y_0)\|$.
7. (S.T. Yau CSMC 2014, team) Let $D \subset \mathbb{R}^n$ be a bounded open set and let $f: \overline{D} \rightarrow \overline{D}$ be a smooth map whose Jacobian determinant is identically 1. Prove that
 - a) for every small ball $B_\varepsilon(x)$ there is a positive integer k with $f^k(B_\varepsilon(x)) \cap B_\varepsilon(x) \neq \emptyset$;
 - b) there exist a point $x \in \overline{D}$ and indices $k_1 < k_2 < \dots$ with $f^{k_j}(x) \rightarrow x$.
8. Prove Hadamard's theorem: a local diffeomorphism $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $\|Df(x)^{-1}\| \leq C$ for all x is a diffeomorphism onto $\mathbb{R}^n$. Show that the bound on Df^{-1} cannot be replaced by non-degeneracy alone.
9. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous with $\|f(x) - f(y)\| \geq \|x - y\|$ for all x, y . Prove that f is injective, that its image is closed, and — using the invariance of domain, which says that an injective continuous map of an open subset of $\mathbb{R}^n$ into $\mathbb{R}^n$ is open — that f is onto. Where exactly does the argument use that the dimensions of the source and the target coincide?

10. (Beckman–Quarles) Let $n \geq 2$ and let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfy $\|f(x) - f(y)\| = 1$ whenever $\|x - y\| = 1$. Prove that f is an isometry. (Start with $n = 2$ and the rhombus built from two unit equilateral triangles.)